MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT S-419()/TCC-73(V)

1. SCOPE

1.1 Scope.- This specification covers the air or vehicular transportable shelter facility part of two versions of Telephone Terminal; the AN/TCC-73()(V)$_1$ or AN/TCC-73()(V)$_2$, which are capable of providing secure multiplex terminal facilities of eight groups of 12 channel PCM multiplexed traffic (96 channels) or four groups of 24 channel multiplex traffic for cable transmission or, with other assemblages, for radio transmission. (See 6.1 and 6.5).

1.2 Description.- Shelter, Electrical Equipment S-419()/TCC-73(V) hereinafter called the shelter facility, includes a modified S-280()/G shelter and all racks, brackets, shelves, tie-downs, electrical components and wiring to enable installation and operation of the assemblage components of the AN/TCC-73()(V)$_1$ or AN/TCC-73()(V)$_2$. (See 6.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)

FSC-5805

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MIL-W-6858	Welding, Resistance, Aluminum Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys and Titanium, Spot and Seam
MIL-T-7928	Terminals, Lug and Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-W-45205	Welding, Gas, Metal-Arc and Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable, for Structures, Excluding Armor
MIL-S-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ELECTRONICS COMMAND

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DL-SC-B-606728	Shelter, Electrical Equipment S-419()/TCC-73(V)
SC-A-46691	RF Cable Assembly Test
SC-D-681007	Signal Assembly, Special Purpose, Electrical
SC-D-681008	Signal Entrance Assembly Wiring Diagram

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish one preproduction sample of the shelter facility for approval. (See 4.3).

3.2 Construction.- The shelter facility shall be constructed in accordance with drawings as listed on DL-SC-B-606728. (See 4.6).

3.3 Electrical installation.-

3.3.1 Power system wiring and components.- Installation of power wires and components (including cable assemblies, receptacles, switches, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps, lighting fixtures, distribution assembly and wiring components constituting a three wire electrical power distribution system) shall be installed and wired in accordance with the drawings. The completed in-place power distribution system shall meet the following requirements:

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3.3.1.1 Continuity of conductors.- Electrical continuity shall exist between each of the pins numbered 1, 2 and 4 (of power connector J1 in power entrance box) and its respective circuit terminations (at receptacles, switches, lighting fixtures and lamp holders) and also between each of the pins numbered 1, 2, and 4 (of power connector J1) and its respective termination (at pins numbered 1, 2 and 4 of power connector J2) when tested in accordance with 4.7.1 and 4.7.1.1.

NOTE. Tests to satisfy the requirements of 3.3.1.1 (See 4.7.1 and 4.7.1.1) shall be accomplished following completion of power system wiring and prior to mounting of AC power receptacles in the wireway.

3.3.1.2 Dielectric strength of insulation.- The insulation of all wires, cables, and components shall withstand a potential of 500 volts DC continuously applied for not less than 30 seconds when tested in accordance with 4.7.1 and 4.7.1.2.

3.3.1.3 Insulation resistance.- The resistance of insulation between any two of the three circuit conductors and also between each of the three circuit conductors and shelter ground shall not be less than 100 megohms when tested in accordance with 4.7.1 and 4.7.1.3.

3.3.2 Power distribution and operational functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as denoted by power system schematic plate when tested in accordance with 4.7.2.

3.3.3 Signal (audio) system wiring and components.- Installation of signal (audio) wires, cables and wiring components (including binding posts, connectors, plugs, jacks, receptacles, switches, terminal blocks, and wiring devices)

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shall be installed and connected in accordance with the drawings. The completed in-place (cables installed) signal (audio) system installation shall meet the following requirements:

3.3.3.1 Continuity and Resistance.- There shall be circuit continuity between both end terminations of all signal connectors and the resistance of the continuous circuit shall not exceed 5 ohms in accordance with 4.7.3.1.

3.3.3.2 Dielectric Strength of Insulation.- The insulation of all wires, cables, and components shall withstand a potential of 500 volts DC continuously applied for not less than 5 seconds when tested in accordance with 4.7.3.2.

3.3.3.3 Insulation Resistance.- The resistance of insulation between each signal line conductor and all other signal line conductor connected to ground shall not be less than 100 Megohms when tested in accordance with 4.7.3.3.

3.3.4 Video system wiring and components.- Installation of video cables and connecting components (including plugs, jacks, connectors, adaptors, fittings, receptacles, and lightning protectors and patch panel shall be in accordance with the drawings. The completed inplace cables installed video system installation shall meet the following requirements:

3.3.4.1 Continuity and circuit resistance.- There shall be circuit continuity between the following video cable terminations (resistance of the continuous circuit shall not exceed 3 ohms when tested in accordance with 4.7.4.1):

a. Both end terminations of the inner conductors.

b. Both end terminations of outer conductor (shield).

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c. Each end of the outer conductor and facility ground terminal.

3.3.4.2 Dielectric strength of insulation.- The insulation of all video cables, components, and lighting protectors shall withstand the following potentials when tested in accordance with 4.7.4.2:

a. 1500 volts DC (applied continuously for not less than 30 seconds) to all video cables terminated at video entrance box with lightning protector elements removed.

b. 1200 volts DC (applied continuously for not less than 30 seconds) to all video cables terminated at video entrance box with lightning protector elements in place.

c. 1500 volts DC (applied continuously for not less than 30 seconds) to all video cable assemblies (to and from) terminated at the video patch panel and TD-204/754 and TD 976 connectors.

d. 1000 volts DC (applied continuously for not less than 30 seconds) to all PCM-TIM video cable assemblies terminated at Video Patch Panel and TD-660, TD 204/754 and TD-976 connectors.

e. 500 volts DC (applied continuously for not less than 10 seconds) to all order wire patch thru cable assemblies terminated at the order wire patch panel.

3.3.4.3 Insulation resistance.- The resistance between each inner conductor and outer conductor (shield) and between each inner conductor and shelter ground shall not be less than 100 megohms when tested in accordance with 4.7.4.3.

3.3.5 Cable Assemblies.- (See 4.4)

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3.3.5.1 Cable assemblies, power and audio frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings and meet the test requirements of MIL-C-3885.

3.3.5.2 Cable assemblies radio frequency.-Radio frequency cable assemblies required for operation of major items of equipment in 6.1 shall be fabricated and tested in accordance with the drawings.

3.3.6 Suppression of radio frequency interference.- The motors in the ventilating equipment used in the facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items specified on the drawings. (See 4.4).

3.4 System operation.- With all operational equipment of the AN/TCC-73$(V)_1$ or $(V)_2$ assemblages installed, and system 1 equipment connected to system 5 equipment, system 2 equipment connected to system 6 equipment, system 3 equipment connected to system 7 equipment and system 4 equipment connected to system 8 equipment connected back to back, communications shall be established between systems. (See 4.8).

3.5 Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicles without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.9).

3.6 Railroad transportation.- The shelter facility shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail. When so loaded, the shelter facility shall be subjected to and withstand the test of paragraph 4.10 without sustaining any damage as mentioned in paragraph 3.5.

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3.7 Drops, flat and rotational.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete. The shelter facility shall be subjected to and withstand the tests of 4.11.1 and 4.11.2 without any damage as mentioned in paragraph 3.5.

3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.12 without any leakage into the facility.

3.9 Fording.- The shelter facility shall be capable of being immersed in water to a depth of 21 inches, measured from the bottom of the skids, for 1 hour without any leakage of water into the facility. (See 4.13).

3.10 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive, and when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.14. There shall be no damage to the insert threads or heads, the insert shall not have rotated from its original position, and there shall be no permanent distortion of the shelter facility mounting surface or insert head or threads. The insert head (if any) shall be properly seated and tight against its bearing surface. Insert threads and outer face of the insert head, shall be free of paint.

3.11 Cleaning.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

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3.12 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the drawings. (See 4.4).

3.13 Marking.-

3.13.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings. (See 4.4).

3.13.2 Nameplate.- Each shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.13.3 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and the assemblage in the three reference planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the centers-of-gravity shall be determined by the contractor using the preproduction sample.

3.13.4 Interior and exterior marking.- All interior and exterior marking of the shelter facility shall be as depicted on the drawings.

3.14 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.15 Seal, vapor.- All joints and raw edges of the framing for the power, signal, video, and antenna entrance boxes, air-conditioner, and exhaust blowers and any other framed openings shall be sealed to prevent entrance of moisture to

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the core material of the shelter panels and to the interior of the shelter. All joints shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be one part Silicone Construction Sealant SE1204 as made by General Electric Company or equal.

3.16 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.16.1 Screws, threaded devices, and related parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

3.17 Soldering.-

3.17.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.17.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.17.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper

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flowing and wetting actions, and shall not be crystallized, over heated or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.18 Tropicalization of material.- Canvas, webbing, wood, etc. shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows:

3.18.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.18.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.18.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2e).

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3.19 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finish weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 Class B and MIL-W-45205 Class B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4).

3.20 Wiring and cabling.- Wiring and cabling shall be in accordance with the drawings and applicable portions of 9 and 19 of MIL-STD-454.

3.20.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.20.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

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3.20.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.20.4 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests Specification MIL-T-7928. (See 4.4 and 4.15).

3.20.5 Identification of wiring.- Wiring shall be identified by colorcoding and wire pair marking, as indicated on the drawings.

3.21 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.16). Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs: (See 4.17).

3.2	Construction
3.3	Electrical installation
3.10	Mounting inserts
3.11	Cleaning
3.12	Finish
3.13	Marking
3.14	Riveting
3.15	Seal, vapor
3.16	Securing of parts
3.17	Soldering
3.18	Tropicalization
3.19	Welding
3.20	Wiring and cabling
3.21	Interchangeability

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commericial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3).

b. Inspection covered by subsidiary documents. (See 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5).

(2) Quality conformance inspection of preparation for delivery. (See 4.18).

4.3 Preproduction inspection.- This inspection shall be performed by the contractor on the preproduction unit unless otherwise specified in the contract. (See 3.1). It shall consist of the preproduction inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (except EVR) (see Inspection Tables I, II and III respectively). Preproduction inspection shall be performed in the following order, unless otherwise specified.

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a. Mounting inserts
b. Electrical installation
c. System operation
d. Truck Transportability
e. Railroad transportability
f. Flat drop
g. Rotational drops
h. Watertightness
i. Fording

The last tests performed shall be a repeat of the Electrical installation tests and System tests. Remaining preproduction tests specified for Groups A, B and C inspections but not included in the above list shall be performed in any order satisfactory to the contracting officer.

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of preproduction inspection and inspection of equipment before preparation for delivery:

Item	Where required
Cable assemblies	3.3.5
Suppression, EMI	3.3.6
Finish	3.12
Marking	3.13
Welding	3.19
Terminals	3.20.4

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the

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Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6). Test equipment for Government verification shall be made available by the contractor.

4.5.1 Group A inspection.- Each unit of equipment shall be inspected for conformance to all inspection and test requirements of Table I of this specification. In addition, equipment conforming to the above requirements shall be verified by a quality audit. The following procedure shall be used for the quality audit:

At the start of this plan 100% of the units shall be inspected until five consecutive units are found free of defects. At that time inspection shall be reduced to one out of five of the units presented. Whenever a defective unit is found, the contractor shall immediately revert to 100% inspection, which shall then be continued until five consecutive units are found free of defects. At that time inspection shall again be reduced to one out of five of the units presented. Samples shall be selected on a random basis.

4.5.1.1 Nonconformance Group A.- A defect of unit of product is any nonconformance with the specified requirement. Any unit of product found defective during inspection shall be rejected. Rejected units may be repaired or corrected and then resubmitted for inspection. The contractor shall simultaneously notify the Procuring Contracting Officer, the government technical activity designated in the contract, and the quality assurance representative and include proposed corrective action whenever the following condition occurs: the contractor's Group A audit fails to clear for sampling, i.e. remains on 100% inspection, for 25 consecutive units. Notification is applicable only to units with major defects.

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Table I - Group A Inspection

Inspection	Reqr Para	Insp Para	
Visual and Mechanical	3.22	4.17	
Electrical			
Power wiring			
Continuity	3.3.1.1	4.7.1.1	All defects are major
Dielectric strength	3.3.1.2	4.7.1.2	
Insulation resistance	3.3.1.3	4.7.1.3	
Power distribution	3.3.2	4.7.2 thru 4.7.2.3	
Signal wiring			
Continuity & circuit resistance	3.3.3.1	4.7.3.1	
Dielectric strength	3.3.3.2	4.7.3.2	
Insulation resistance	3.3.3.3	4.7.3.3	
Video wiring			
Continuity & circuit resistance	3.3.4.1	4.7.4.1	
Dielectric strength	3.3.4.2	4.7.4.2	
Insulation resistance	3.3.4.3	4.7.4.3	

NOTE: The order of inspection of Group A with respect to Groups B and C shall conform to the order specified in 4.3.

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4.5.2 Group B inspection.- Equipment shall be assembled in discrete lots for quality conformance inspection, utilizing the procedures of MIL-STD-105 for lot formation. Each lot shall consist of units produced within a calendar month. This inspection, including sampling, shall conform to Table II and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on units of production that have passed Group A inspection except for Mounting Inserts and Interchanbeability Inspection which are performed prior. The government reserves the right to require complete retest of any sample which is reworked after failing Group B inspection.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in Table II shall be as follows:

AQL	Inspection level
4.0%	S-4
2.5%	S-4

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed on the same equipment in the order in which they appear.

Table II - Group B Inspection

Inspection	Reqr Para	Insp Para	AQL % Def (see 4.5.2.1)
Mounting inserts	3.10	4.14.2	4.0 (AOQL-5.0)
Interchanbeability	3.21	4.16	4.0 (AOQL-5.0)
Watertightness	3.8	4.12	2.5 (AOQL-3.0)
Fording	3.9	4.13	2.5 (AOQL-3.0)

NOTE: The order of inspection of Group B with respect to Groups A and C shall conform to the order specified in 4.3. The order of inspection within Group B shall be as in Table II.

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4.5.2.3 Alternate Group B inspection procedures.- Contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plan stated above. If this alternate is selected the procedure of implementation must be in accordance with MIL-STD-1235, "Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes" and the Group B sampling plans for the AOQL's listed in Table II of this specification shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=½. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract. Before starting production Contractor shall notify the contracting officer in writing which method he intends to use. The method may not be changed during production without written approval of the government.

4.5.3 Group C inspection.- This inspection shall be as listed in Table III.

Table III - Group C Inspection

Inspection	Reqr Para	Insp Para
Solderless-terminal-lug connection	3.20.4	4.15
Construction verification	3.2	4.6
Mounting inserts	3.10	4.14.1
System operation	3.4	4.8
Truck Transportability	3.5	4.9
Railroad transportability	3.6	4.10
Drops, Flat and Rotational	3.7	4.11

NOTE: The order of inspection of Group C with respect to Groups A and B shall conform to the order specified in 4.3. The order of inspection within Group C for all inspections listed shall be as listed in Table III. Units subjected to Group C Drop Tests shall subsequently be subjected to and meet the watertightness and fording tests of 4.12 and 4.13.

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4.5.3.1 Sampling for inspection of construction evaluation.- The Equipment Verification Review, in accordance with paragraph 4.6 shall be performed once on each part and assembly, randomly selected, and on their installation into one of first 10 shelter facilities produced. This inspection shall not be performed on the preproduction unit. This inspection shall be performed as the selected shelter is being assembled.

4.5.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Sampling for inspection of Drop Tests.- One sample, representative of the first 50 units or fraction thereof produced, shall be subjected to the drop tests. This sample shall be the same unit previously subjected to the truck and railroad transportation tests of 4.5.3.4. Thereafter, one sample from each additional 150 units or fraction thereof produced shall be subjected to the drop tests.

4.5.3.4 Sampling for other inspection.- For inspection tests listed below, one unit shall be selected at random from the first 10 units of fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the government.

a. Mounting inserts
b. System operation
c. Truck transportability
d. Railroad transportability

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no permanent deformation or cracking.

4.5.3.5 Noncompliance.- All quality conformance inspection shall be halted, including group A and group B inspections, upon the occurrence of any group C failure. The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on the processes and materials, as applicable, and on all units of product which are manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract with a copy to the Government procurement quality assurance representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the technical activity) and group A and group B inspections may be reinstituted. However, final acceptance and shipment will be withheld until group C reinspection results have shown that the corrective action was effective and approved by the technical activity. (See "Delivery and Conditional Acceptance of Contract Items Subject to Group C Inspection" provisions elsewhere in the contract.

4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and pass Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage (See 6.4).

4.6 Construction verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings (see 3.2) cited in the Technical Data Package (as supplemented by all approved engineering changes). EVR shall

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a. An audit to establish that "as-built" shelter facilities, including individual parts, their use in sub-assemblies, and assemblies (components), and their final installation into the shelter forming the shelter facility, are in accordance with the end product drawings.

b. An audit to establish that "as-built" shelter facilities, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each shelter part, components, and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information:

(1) Identification of parts, components, shelter facilities and details proving compliance with 4.6a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

d. Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the government technical organization cited in the contract. The results of the EVR shall be subject to verification by authorized government personnel at the time the first unit or production lot is offered for acceptance. Government verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to government personnel during the verification. The contractor shall provide the following:

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(1) Segregation of parts, components of shelter facilities, to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit government review of the shelter facility, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

4.7 Electrical installation tests.- The tests of 4.7.1 through 4.7.4.3a shall be performed on each shelter facility prior to installation of other major components comprising the AN/TCC-73 or dummy loads. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to the shelter facility shall be 115 VAC, 60 cycles, single phase.

4.7.1 Power system and components installation tests.- Prior to conducting power wiring and components tests of 4.7.1.1 and 4.7.1.2, and 4.7.1.3, the following shall be accomplished: (See power schematic plate).

a. Remove or disconnect the following electrical loads:

(1) Fluorescent lamps and lamp starters.

(2) Incandescent cold start lamps.

(3) Power distribution boxes neon indicator lamps.

(4) Power indicator neon lamp.

(5) Blower-1 and blower-2.

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b. Remove power distribution box cover panel and disconnect both conductors from the voltmeter terminals. Position or isolate disconnected leads so that they do not short to each other or to shelter ground.

c. Plug in cable from rack equipment raceway to associated receptacle.

4.7.1.1 Continuity of conductors.- Continuity measurements which satisfy the requirements of 3.3.1.1 shall be made on each power-line circuit as indicated by the following paragraphs using an aural or visual indicating continuity test instrument and suitable test jig. The instrument shall produce a test current between 0.2 and 2.0 amperes AC when circuit continuity exists.

a. Remove and extend all connected power receptacles from wireway so that the outlet mounting brackets do not make electrical contact to the wireway. (See Figure 1)

b. Place the following circuit controls in the ON position:

(1) Fluorescent lamp switches.

(2) Incandescent cold start switches.

(3) Bypass blackout switch (by pass).

(4) Main circuit breaker.

(5) All branch circuit breakers.

(6) Blower switches.

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c. Measure continuity from Pin 1 (of power connector IN(J1) power entrance box) to the following: brass contact on each receptacle, black wire termination at each fluorescent lamp; and black wire termination at each incandescent lamp.

d. Place all controls listed under procedure 4.7.1.1b in OFF position, except main circuit breaker.

e. Measure continuity from Pin 2 (of power connector J1) to the following: silver contacts on each receptacle; white wire termination at each fluorescent lamp and white wire termination at each incandescent lamp.

f. Place main circuit breaker in OFF position.

g. Measure continuity from Pin 4 (ground) (of power connector J1) to each ground terminal on receptacles.

h. Measure continuity from Pin 4 (ground) (of power connector J1) to shelter ground terminal located in power entrance box panel.

i. Measure continuity from Pin 1 of IN connector J1 to Pin 1 of OUT connector; J2 from Pin 2 of IN connector J1 to Pin 2 of OUT connector J2; and from Pin 4 of IN connector J1 to Pin 4 of OUT connector J2.

j. Replace and secure power receptacles removed by procedure 4.7.1.1a.

k. Measurements made by 4.6.1.1c, e, g, h and i shall indicate existence of electrical circuit continuity. (See 3.3.1.1)

4.7.1.2 Dielectric strength of insulation.- Dielectric strength of insulation tests which satisfy the requirements

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of 3.3.1.2 shall be made on each power-line circuit indicated by the following paragraphs (Method 301 of MIL-STD-202, shall be employed using a nondestructive DC dielectric tester):

a. Place all circuit controls listed in procedure 4.7.1.1b in the ON position, except fluorescent lamp swtiches.

b. Connect tester between Pin 1 and Pin 4 of power IN connector in power entrance box. Adjust tester for a value of 500 volts DC and maintain this potential for a period of 30 seconds. The requirements of 3.3.1.2 shall be met.

c. Place fluorescent lamp switches in the ON position.

d. Repeat procedure 4.7.1.2b with tester connected between Pin 1 and Pin 2 of power IN connector.

e. Repeat procedure 4.7.1.2b with tester connected between Pin 2 and Pin 4 of power IN connector.

NOTE: To eliminate danger of shock.- Discharge conductors following each test.

4.7.1.3 Insulation resistance.- Insulation resistance tests which satisfy the requirements of 3.3.1.3, shall be made on each power-line circuit indicated by the following paragraph (Method 302 of MIL-STD-202 shall be employed using a megohmmeter or equivalent tester).

a. Connect tester between Pin 1 and Pin 2 of power connector in power entrance box. Adjust tester for a value of not less than 100 nor more than 500 volts DC. Maintain this potential for 1 minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.3.1.3 shall be met.

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b. Repeat procedure 4.7.1.3a with tester connected between Pin 1 and Pin 4 of power OUT connector.

c. Repeat procedure 4.7.1.3a with tester connected between Pin 2 and Pin 4 of Power IN connector.

NOTE: To eliminiate danger of shock. Discharge conductors following each test.

4.7.2 Power distribution and operational functions tests.- Power distribution and operational functions tests which satisfy the requirements of 3.3.2 shall be made as indicated by the following paragraphs:

4.7.2.1 Test preparation.- (See power schematic plate).

a. Replace or reconnect all electrical loads and line voltmeter removed or disconnected by procedure 4.7.1a and b.

b. Replace power distribution box cover.

c. Place circuit breakers and switches in the OFF position.

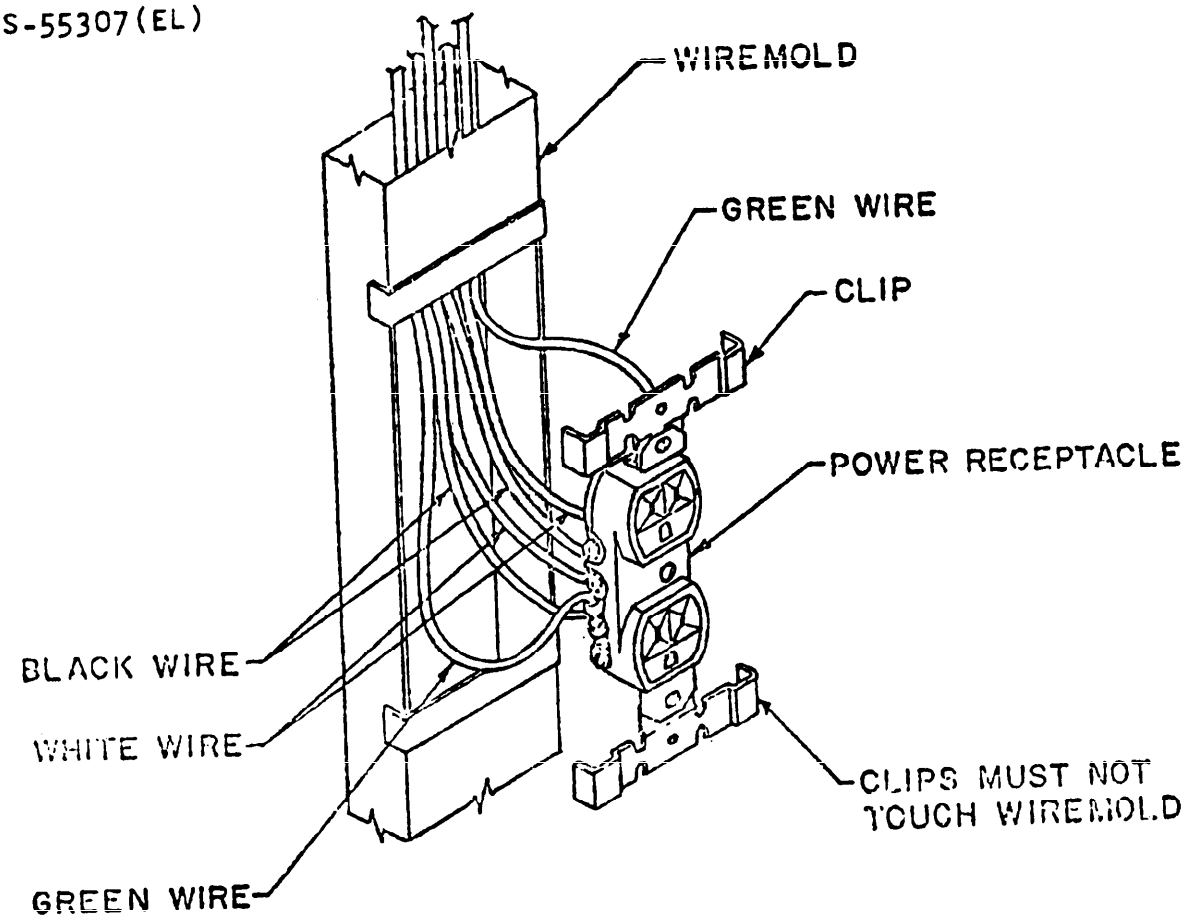
d. Place a 115 volt AC test lamp with grounding type plug into each convenience and equipment receptacle.

e. Open outer exhaust port covers of blowers 1 and 2.

f. Place by-pass blackout switch in the by-pass position.

g. Insert AC line plugs of exhaust blowers (1 and 2) into receptacles.

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POWER RECEPTACLE REMOVED & EXTENDED FROM WIREMOLD

FIGURE NO. 1

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4.7.2.2 Line voltmeter and indicator lamp.-

a. Apply 115 volt AC (nominal), 60 cycles, power to power connector J1 in power entrance box. Power indicator neon lamp shall light; test lamp in receptacle shall light; and facility voltmeter in power distribution panel shall indicate power to the shelter.

b. Measure specific line voltage by connecting a calibrated external AC voltmeter between Pin 1 and Pin 2 on power connector J2.

c. Read voltage indicated on facility voltmeter. The voltage read shall be within 3% of the voltage read by 4.7.2.2b.

d. Place main circuit breaker in ON position.

4.7.2.3 Circuit breakers and switches.-

Circuit #1.- (Lights circuit breaker CB1).

a. Place circuit breaker in the ON position. Neon lamp above CB1 shall light.

b. Place fluorescent light switches in ON position. All fluorescent lamps shall light.

c. Place incandescent light switches in ON position. Incandescent lamps shall light.

d. Place circuit breaker in OFF position. Neon lamp and all fluorescent and incandescent lamps shall go out.

e. Place circuit breaker in ON position. With shelter door open, place by-pass blackout switch in black-out position. All fluorescent and incandescent lamps shall go out.

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f. Close shelter door. All fluorescent and incandescent lamps shall light.

g. Open door and place by-pass blackout switch in by-pass position.

Circuit No. 2 (Blowers circuit breaker CB2)

a. Place circuit breaker in ON position. Neon lamp above CB2 shall light.

b. Place blower 1 switch in ON position. Blower 1 shall operate.

c. Place blower 2 switch in ON position. Blower 2 shall operate.

d. Place circuit breaker in OFF position. Neon lamp shall go out and blowers 1 and 2 shall cease to operate.

Circuit No. 3.- (Heater #1 circuit breaker CB3).

a. Place circuit breaker in ON position. Neon lamp shall light, and test lamp in receptacle shall light.

b. Place circuit breaker in the OFF position. Neon lamp shall go out and test lamp in receptacle shall go out.

Circuit No. 4 (Equipment #1 Circuit Breaker CB-4):

a. Place circuit breaker in ON position. Neon lamp above CB-4 shall light, and test lamp in receptacles for equipment CV-1548 Sys 1-4, TD-660 Sys 1-4, KG Sys 1-4, TD-204/754 Sys 1-2 and TD-976-1 shall light.

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b. Place circuit breaker in OFF position. Neon lamp shall go out and test lamps in receptacles for equipment CV-1548, Sys 1-4, TD-660 Sys 1-4, KG Sys 1-4, TD-204/754 Sys 1-2 and TD-976-1 shall go out.

Circuit No. 5 (Convenience & Intercom Circuit Breaker CB-5):

a. Place circuit breaker in ON position. Neon lamp above CB-5 shall light and test lamp in each receptacle shall light.

b. Place circuit breaker in OFF position. Neon lamp shall light and test lamp in receptacles shall go out.

Circuit No. 6 (Spare Circuit Breaker CB-6):

a. Place circuit breaker in ON position. Neon lamp above CB-6 shall light.

b. Place circuit breaker in OFF position. Neon lamp shall go out.

Circuit No. 7 (Heater #2 Circuit Breaker CB-7):

a. Place circuit breaker in ON position. Neon lamp above CB-7 shall light, and test lamp in receptacle shall light.

b. Place circuit breaker in the OFF position. Neon lamp shall go out and test lamp in receptacle shall go out.

Circuit No. 8 (Equipment #2 Circuit Breaker CB-8):

a. Place circuit breaker in ON position. Neon lamp above CB-8 shall light, and test lamp in receptacles for equipment CV-1548 Sys 5-8, TD-660 Sys 5-8, TD-204/754 Sys 3-4

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b. Place circuit breaker in OFF position. Neon lamp shall go out and test lamps in receptacles for equipment CV-1548 Sys 5-8, TD-660 Sys 5-8, TD-204/754 Sys 3-4 and TD-976-2 shall go out.

Circuit No. 9 (Main Circuit Breaker CB-9):

a. Place all circuit breakers in ON position. All neon lamps shall light, Blower 1 and 2 shall operate, test lamp in each receptacle shall light.

b. Place main circuit breaker in OFF position. All neon lamps shall go out. Blower 1 and Blower 2 shall cease to operate. Test lamp in each receptacle shall go out.

Circuit No. 10 (Outside Convenience Circuit Breaker CB-10):

a. Place circuit breaker in ON position (IN). Test lamp in receptacle shall light.

b. Place circuit breaker in OFF position (OUT). Test lamp in receptacle shall go out.

4.7.2.4 For remainder of tests, place main circuit breaker, and light circuit breakers in the ON position and all other circuit breakers in the OFF position.

4.7.3 Signal (audio) system wiring and components installation tests.- Reference drawings SC-D-681007 and SC-D-681008.

4.7.3.1 Continuity and circuit resistance.- Continuity resistance which satisfy the requirements of 3.3.3.1 shall be made on each audio circuit indicated by the following paragraphs (using a low resistance continuity tester or equivalent producing a test current of not more than 0.010 amps.)

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a. Prior to conducting signal wiring and components tests of 4.7.3.1, 4.7.3.2 and 4.7.3.3 all system cables shall be plugged into respective connectors on signal entrance boxes by systems and channels.

b. Place all cable/binding post channel switches, S1 through S12 of system 1 through 8, in BINDING POST position.

c. Connect tester between Pin A (of audio connector, equipment end system 1, channels 1-4) and Pin 2B (of audio connector system 1 in signal entrance box) (see Drawings SC-D-681007 and SC-D-681008). No circuit continuity shall exist.

d. Place channel switch S1 of system 1 to CABLE position. Circuit continuity shall exist and meet the requirements of 3.3.3.1. Replace channel switch S1 to BINDING POST position.

e. Repeat procedures 4.7.3.1c through d for each remaining conductor (Pin B through Pin S of audio connector to equipment end) its corresponding pin of audio connector system 1; each time placing to CABLE position the channel switch correlating with the circuit being tested and replacing to BINDING POST position following completion of test.

f. Repeat procedures 4.7.3.1c through e for each conductor terminated at audio connectors of system 1 channels 5-12.

g. Repeat procedures 4.7.3.1b through f for each conductor terminated in audio connector system 2 through 8.

h. Place all CABLE/BINDING POST channel switches, S1 through S12 of system 1 through 8, to CABLE position.

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i. Connect tester between Pin A of audio connector and binding post 2B system 1, in signal entrance box. No circuit continuity shall exist.

j. Place channel switch S1 of system 1 to BINDING POST position. Circuit continuity shall exist and meet the requirements of 3.3.3.1. Replace channel switch S1 to CABLE position.

k. Repeat procedure 4.7.3.1i and j for each of the remaining conductor (Pin B through Pin S of audio connector equipment end) to its corresponding binding post of system 1; each time placing to BINDING POST position the channel switch correlating with the circuit being tested and replacing to CABLE position following test.

l. Repeat procedures 4.7.3.1i through k for each conductor terminated at audio connectors of system 1 channel 5-12.

m. Repeat procedures 4.7.3.1i through l for each conductor terminated on audio connectors system 2 through 8.

n. Connect phone/intercom parallel interconnecting cable assembly between the phone/intercom plugs located on raceway and signal entrance boxes.

o. Circuit continuity that meets the requirements of 3.3.3.1 shall exist between the following points:

From black conductor of telephone cable to:
Pin 25A of audio connector system 1 through system 8.
Binding Post 25A of system 1 through system 8.
Phone A binding post at power entrance box.

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From red conductor of telephone cable to:
Pin 25B of audio connector system 1 through system 8.
Binding Post 25B of system 1 through system 8.
Phone B binding post at power entrance box.

From black conductor of intercom cable to:
Pin 26A of audio connector system 1 through system 8.
Binding Post 26A of system 1 through system 8.
Intercom A binding post at power entrance box.

From red conductor of intercom cable to:
Pin 26B of audio connector system 1 through system 8.
Binding Post 26B of system 1 through system 8.
Intercom B Binding Post at power entrance box.

p. Disconnect phone/intercom cable installed in 4.7.3.1n above from raceway only.

4.7.3.2 Dielectric strength of insulation.- Dielectric strength of insulation tests which satisfy the requirements of 3.3.3.2 shall be made for each audio circuit indicated by the following paragraphs (Method 301 of MIL-STD-202 shall be used utilizing a non-destructive dielectric test instrument and suitable test jigs or an automatic tester.

a. Place all CABLE/BINDING POST channel switches, S1 through S12 of system 1 through 8 in the CABLE position.

b. Connect the following to facility ground terminal:

(1) Pins 1A through 26B of audio connector system 1 through system 8.

(2) Binding posts 1A through 26B of system 1 through system 8.

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c. Connect tester between the facility ground terminal and Pin A of audio connector equipment end channels 1 through 4. Remove ground connection from Pin 2B of audio connector system 1. (See Drawings SC-D-681007 and SC-D-681008).

d. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.3.2 shall be met.

e. Repeat procedures 4.7.3.2c through f for each remaining conductor (Pin B through Pin S of audio connector channels 1 through 4 each time ungrounding the pin of audio connector system 1 correlating with the circuit being tested and reconnecting pin to facility ground terminal following completion of the test.

f. Repeat procedures 4.7.3.2c through e for each conductor terminated on audio connectors of system 1 channel 5-12.

g. Repeat procedures 4.7.3.2c through f for systems 1 through 8.

h. Place all CABLE/BINDING POST channel switches, S1 through S12 of system 1 through 8 in the BINDING POST position.

i. Repeat procedures 4.7.3.2c through g substituting BINDING POST of system 1 through 8 for audio connector system 1 through system 2 respectively.

j. In power entrance box connect phone and intercom BINDING POST to facility ground terminal.

k. Remove ground from BINDING POST A of phone circuit and connect tester between facility ground terminal and BINDING POST A.

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1. Adjust tester for 200 VDC output and maintain this potential for a period of not less than 5 seconds. The requirements of 3.3.3.2 shall be met.

m. Repeat procedures 4.7.3.2 k and l for each remaining conductor, each time ungrounding BINDING POST correlating with circuit being tested and reconnecting BINDING POST to facility ground terminal following completion of test.

NOTE. To eliminate danger of shock. Discharge conductors following each test.

4.7.3.3 Insulation resistance.- Insulation resistance which satisfies the requirements of 3.3.3.3 shall be made on each audio circuit indicated by the following paragraphs (Method 302 of MIL-STD-202 shall be used, utilizing a megohmmeter or equivalent tester, and suitable test jig or an automatic tester).

a. Repeat procedures 4.7.3.2a through i, except 4.7.3.2d which shall be replaced by 4.7.3.3b below:

b. Adjust tester to 500 volts DC output. The requirements of 3.3.3.3 shall be met.

NOTE. To eliminate danger of shock. Discharge conductors following each test.

4.7.4 Video system wiring and components installation.-

4.7.4.1 Continuity and circuit resistance.- Continuity resistance which satisfies the requirements of 3.3.4.1 shall be made on each video circuit indicated by the following paragagraph (using a low resistance continuity tester):

a. Circuit continuity resistance shall be tested at video patch panel. Method of testing shall be

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inner conductor to inner conductor and outer conductor (shield) to outer conductor (shield) in accordance with Table IV.

Table IV - Continuity Measurements

From Equipment Cable	To Video Patch Panel
TD-660 - SYS-1, PCM OUT	TD-660 - SYS-1, PCM OUT
TD-660 - SYS-1, PCM IN	TD-660 - SYS-1, PCM IN
TD-660 - SYS-1, TIM IN	TD-660 - SYS-1, TIM IN
TD-660 - SYS-1, TIM OUT	TD-660 - SYS-1, TIM OUT
TD-660 - SYS-1, ACT PCM OUT	TD-660 - SYS-1, ACT PCM OUT
TD-660 - SYS-1, SYNC IN	TD-660 - SYS-1, SYNC IN
TD-660 - SYS-1, SYNC OUT REC	TD-660 - SYS-1, SYNC OUT REC
TD-660 - SYS-1, SYNC OUT XMTR	TD-660 - SYS-1, SYNC OUT XMTR
Repeat above for TD-660 SYS 2 thru 8	Repeat above for TD-660 SYS 2 thru 8
TD-204/754 SYS-1, from cable	TD-204/754 SYS-1 from cable
TD-204/754 SYS-1, to cable	TD-204/754 SYS-1 to cable
TD-204/754 SYS-1, PCM OUT-1	TD-204/754 SYS-1 PCM OUT-1
TD-204/754 SYS-1, PCM IN-1	TD-204/754 SYS-1 PCM IN-1

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TD-204/754 SYS-1, TIM IN	TD-204/754 SYS-1, TIM IN
TD-204/754 SYS-1, TIM OUT-1	TD-204/754 SYS-1, TIM OUT-1
TD-204/754 SYS-1, PCM OUT-2	TD-204/754 SYS-1, PCM OUT-2
TD-204/754 SYS-1, PCM IN-2	TD-204/754 SYS-1, PCM IN-2
TD-204/754 SYS-1, TIM OUT-2	TD-204/754 SYS-1, TIM OUT-2
Repeat above for TD-204/754 SYS-2 thru 4	Repeat above for TD-204/754 SYS-2 thru 4
TD-204/754 SYS-1 Patch Thru Pin A	TD-204/754 SYS-1 Patch Thru Pin A
TD-204/754 SYS-1 Patch Thru Pin B	TD-204/754 SYS-1 Patch Thru Pin B
TD-204/754 SYS-1 Patch Thru Pin C	TD-204/754 SYS-1 Patch Thru Pin C
TD-204/754 SYS-1 Patch Thru Pin D	TD-204/754 SYS-1 Patch Thru Pin D
TD-204/754 SYS-1 Patch Thru Pin E	TD-204/754 SYS-1 Patch Thru Pin E
TD-204/754 SYS-1 Patch Thru Pin F	TD-204/754 SYS-1 Patch Thru Pin F
TD-204/754 SYS-1 Patch Thru Pin G	TD-204/754 SYS-1 Patch Thru Pin G
TD-204/754 SYS-1 Patch Thru Pin H	TD-204/754 SYS-1 Patch Thru Pin H
Repeat above for TD-204/754 SYS 2 thru 4	Repeat above for TD-204/754 SYS 2 thru 4
TD-976, #1, PCM OUT SYS-1	TD-976, #1 PCM OUT SYS-1
TD-976, #1, PCM IN SYS-1	TD-976, #1 PCM IN SYS-1
TD-976, #1, TIM OUT SYS-1	TD-976, #1 TIM OUT SYS-1
Repeat above for Systems 2 thru 8	Repeat above for Systems 2 thru 8

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TD-976, #2, PCM OUT SYS-1	TD-976, #2, PCM OUT SYS-1
TD-976, #2, PCM IN SYS-1	TD-976, #2, PCM IN SYS-1
TD-976, #2, TIM OUT SYS-1	TD-976, #2, TIM OUT SYS-1
Repeat above for Systems 2 thru 8	Repeat above for Systems 2 thru 8

FROM Video Ent Box	TO Video Patch Panel TD-204/754
System 1 (Male)	FROM Cable SYS-1
System 1 (FEMALE)	TO Cable SYS-1
Repeat above for Systems 2 thru 4	Repeat above for Systems 2 thru 4
System 1-8 (MALE)	TD-976 From Cable SYS 1-8
System 1-8 (FEMALE)	TD-976 to Cable SYS 1-8
Repeat above for TD-976 #2	Repeat above for TD-976 #2

4.7.4.2 Dielectric strength of insulation.- Dielectric strength of insulation tests which satisfy the requirements of 3.3.4.2 shall be made on each video circuit indicated by the following paragraphs (Method 301 of MIL-STD-202 shall be used utilizing a nondestructive dielectric tester):

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a. Position or separate the unconnected video connectors so that none are shorted to each other or to facility ground.

b. Remove all lightning protector elements.

c. Apply test potential of 1500 volts for not less than 30 seconds, between TO (Male) coaxial connector system 1 and terminal ground in video entrance box. The requirements of 3.3.4.2 shall be met.

d. Repeat procedure 4.7.4.2c for from (female) coaxial connector system 1.

e. Repeat procedure 4.7.4.2c and d for coaxial connectors system 2 thru 4 and system (1-8).

f. Replace lightning protector elements at video entrance box.

g. Repeat procedure 4.7.4.2c thru e applying a test potential of 1200 volts DC. The requirements of 3.3.4.2 shall be met.

h. Apply a test potential of 1000 V DC for not less than 30 seconds between inner conductor and shield of connector at the video patch panel for each TD-204/754, TD-660 and TD-976. The requirements of 3.3.4.2d shall be met.

i. Apply test potential of 500 V DC for not less than 10 seconds between each pin indicated below and all other pins connected to shield of each patch thru connector at order wire patch panel. The requirements of 3.3.4.2 shall be met:

(1) Order Wire Patch-Thru Systems 1 thru 4 Pins A, B, C, D, E, F, G and H.

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4.7.4.3 Insulation Resistance.- Insulation resistance tests which meet the requirements of 3.3.4.3 shall be made on each video circuit indicated by the following paragraphs (method 302 of MIL-STD-202 shall be used utilizing a megometer or equivalent tester).

a. Repeat procedure 4.7.4.2a thru i except that the test potential shall be not less than 100 nor more than 500 V DC output applied unitl the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance.

4.8 Systems operational tests.- Install operational equipments in their respective rack locations and connect all power, signal and video cable assemblies to their designated receptacles and connectors. (See 3.4).

4.8.1 Electrical heaters operational tests.-

a. Place heater-1 in the FAN position.

b. Place heater-1 circuit breaker CB3 in the ON position. Heater-1 fan shall operate and blow air out through louvers.

c. Place thermostat control on heater-1 to a high heating position. Place heater-1 in the HEAT position. Shelter facility ammeter shall indicate approximately 15 amps. After a few minutes the heater shall blow warm air. Check that the thermostat controls heater operation by turning thermostat to a lower temperature until a click is heard. The ammeter shall show a drop of about 13 amps. The heater's fan shall continue to operate. Place the heater in the off position. The heaters fan shall cease to operate.

d. Repeat steps a through c above for heater-2 and circuit breaker CB7.

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4.8.2 Equipment operational tests.-(See Figure 2 for block diagram).

a. At video patch panel make the following cable connections using video patch cords:

TD-660-SYS-1	TD-204 or TD-754 SYS-1
PCM OUT	PCM IN-1
PCM IN	PCM OUT-1
TIM OUT	TIM IN
TIM IN	TIM OUT-1
TD-660 SYS-2	**TD-204 or TD-754 SYS-1**
PCM OUT	PCM IN-2
PCM IN	PCM OUT-2
TIM IN	TIM OUT-2
TD-660 SYS-3	**TD-204 or TD-754 SYS-2**
PCM OUT	PCM IN-1
PCM IN	PCM OUT-1
TIM OUT	TIM IN
TIM IN	TIM OUT-1
TD-660 SYS-4	**TD-204 or TD-754 SYS-2**
PCM OUT	PCM IN-2

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PCM IN — PCM OUT-2

TIM IN — TIM OUT-2

TD-660 SYS-1 SYNC-OUT XMTR TO

TD-660 SYS-2 SYNC-IN

TD-660 SYS-3 SYNC-OUT XMTR TO

TD-660 SYS-4 SYNC-IN

TD-204 or TD-754 SYS-1	VID ENT SYS-1
TO CABLE	TO CABLE
FROM CABLE	FROM CABLE
TD-204 or TD-754 SYS-2	VID ENT SYS-2
TO CABLE	TO CABLE
FROM CABLE	FROM CABLE

b. Interconnect system 1 and system 2 video connectors at video entrance box using Cable adapters.

c. Place all CABLE/BINDING POST channel switches, S1 thru S12 of systems 1 thru 8 in BINDING POST position at signal entrance boxes.

d. Perform the following checks on the Converter Telephone Signal CV-1548( )/G:

(1) Check to see that the power switch is operated to OFF.

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(2) Operate the CH-2W-4W switch to 2W and the signaling mode switches to AC.

(3) Operate the power switch to ON.

(4) Check that the 20 cycle indicator on panel 18A2 is glowing).

(5) Operate the meter selector switch through -, +, 20 cycle drive, and 20 cycles, and check for a yellow area indication on the TEST ALIGN meter.

(6) Operate the meter selector switch to 1600 cycles and check for a green area indication on the TEST ALIGN meter. If necessay, adjust the ADJ 1600 control on panel 18A2 for a center indication in the green area.

e. Perform the following procedures on each TD-660( )/G:

(1) Check to see that the AC power switch is operated to OFF.

(2) Operate switches I, II, III and IV to OFF position.

(3) Operate the AUX switch to OUT.

(4) Operate the master-slave switch to master positions for systems 1, 3, 5, and 7 and slave po-stion for systems 2, 4, 6 and 8.

(5) Operate 2W-4W switch to 2W position.

(6) Operate 6CH/12CH mode switch to 12CH.

(7) Operate the AC power switch to ON. The power indicator lamp should light and the test align meter should indicate in the green area.

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(8) Operate the selector switch I to each of the positions listed below and observe the test align meter:

Switch Position	Meter Indication
+7	Yellow
+12	Hairline
+4	Green
-12	Green
-6	Green
-4	Green

(9) Operate selector switch I to SW III position.

(10) Operate selector switch III to OSC postion. The test align meter should indicate in the green area.

f. Perform the following procedures on each TD-204/U or TD-754 connected in system, where applicable:

(1) Operate the AC POWER, and TALK-OFF-SIG switches to OFF.

(2) Depress the PUSH TO RELEASE CHASSIS button and slide the TD-204/U out of its case far enough to expose the service facility panel and panel 6A4.

(3) Operate the TRAFFIC SEL switch to 24 channel position.

(4) Operate NORM OPR-ZERO SET-READ switch to NORM OPR.

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(5) Operate both MILES switches to 0.

(6) Operate the TONE-OFF switch on panel 6A2 to OFF.

(7) Operate the MILE switch on panel 6A4 to ½ and on panel 6A5 to ½.

(8) Operate the AC POWER switch to ON. Check to see that the AC POWER and ALARMS NO CABLE CURRENT indicators light and the buzzer sounds. Silence the ALARMS BUZZER OFF switch.

(9) Operate the CABLE POWER switch to OFF.

(10) Operate the METER SELECT switch to SERV FAC.

(11) Operate SERV SEL switch -10, +10, sum ±3 and bal, and check for a yellow area indication on the test align meter for each position. Operate SERV SEL switch to +10, return TD-204 to case.

g. Connect a Telephone Set TA-312( )/PT phone 1 to BINDING POST pair 2WT/R channel 1 system 1, signal entrance box roadside, (rear wall).

h. Connect a Telephone Set TA-312( )/PT phone 2 to BINDING POST pair 2WT/R channel 1 system 3, signal entrnace box roadside (rear wall).

i. Ring phone 2 from phone 1. Ring phone 1 from phone 2. Establish communications between phones.

j. Repeat step 4.8.2.i above with phones connected to 2WT/R for the remaining channels of systems 1 and 3.

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k. Set 2 wire-4 switches in Converter Telephone Signal CV-1548( )/U and Multiplexer TD-204( )/U or TD-754( )/U.

l. Connect Phone 1 to BINDING POST pair 2WT/R channel 1,System 1. Connect Phone 2 to BINDING POST Pair 4WT, channel 1, system 3. Receive communications at Phone 1 from Phone 2.

m. Connect Phone 1 to BINDING POST pair 2WT/R channel 1, system 3. Receive communications at Phone 2 from Phone 1.

n. Repeat 4.8.2(1)(m) above for remaining channels of system 1 and 3.

o. Repeat 4.8.2g thru m for system 2 and system 4.

p. Repeat 4.8.2a substituting the following patching:

TD-660( )/U SYS-5 for TD-660( )/U SYS-1
TD-660( )/U SYS-6 for TD-660( )/U SYS-2
TD-660( )/U SYS-7 for TD-660( )/U SYS-3
TD-660( )/U SYS-8 for TD-660( )/U SYS-4
TD-204/754 SYS-3 for TD-204/754 SYS-1
TD-204/754 SYS-4 for TD-204/754 SYS-2

q. Repeat 4.8.2b substituting system 3 for 1 and system 4 for 2.

r. Repeat 4.8.2(d) thru (o) substituting system 5 for system 1, system 6 for system 2, system 7 for system 3 and system 8 for system 4.

s. Turn off all equipment and disconnect all cable connections and secure operational test.

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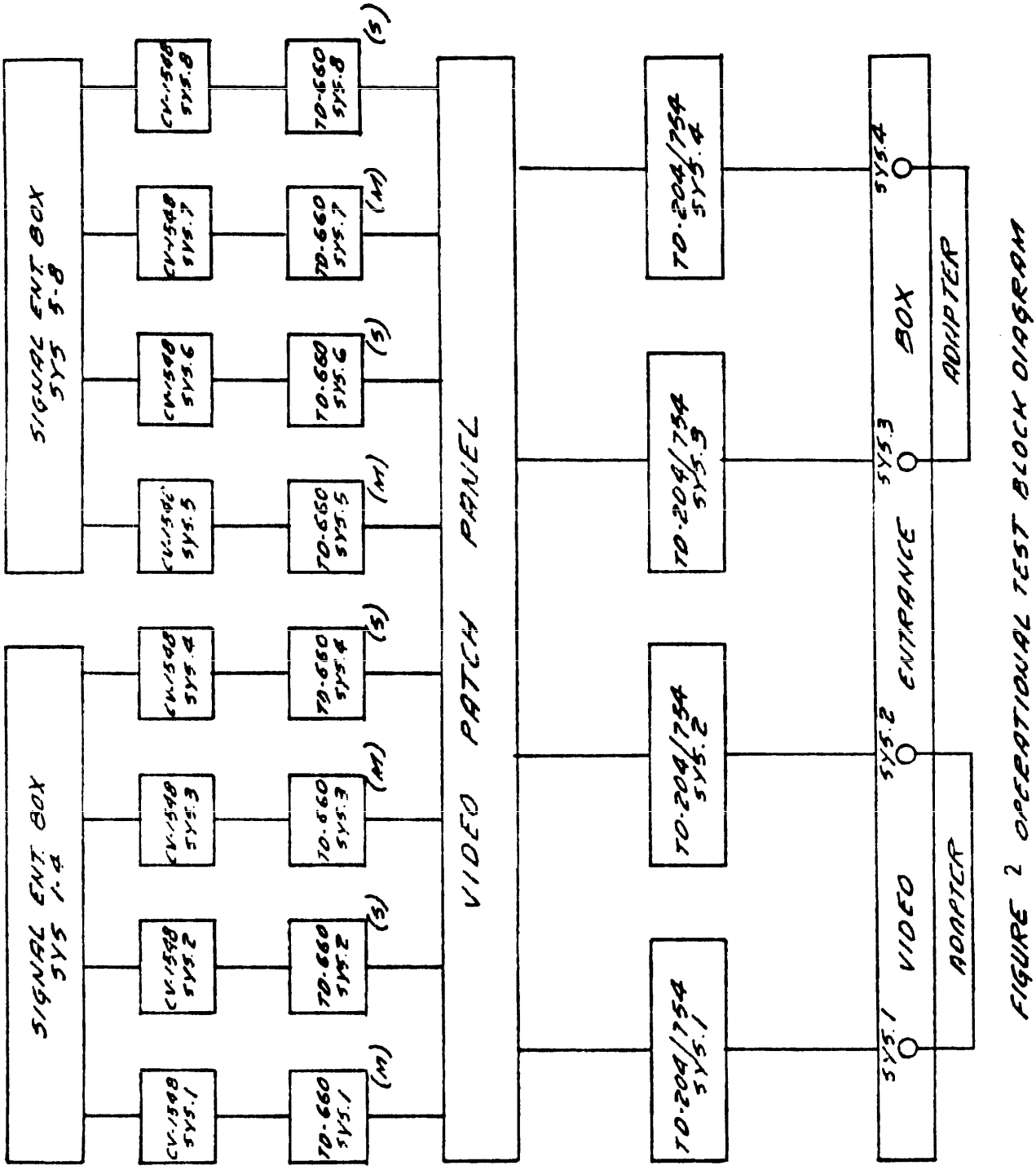


FIGURE 2 OPERATIONAL TEST BLOCK DIAGRAM

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4.9 Vehicular transportation test.- The facility shall be secured (and blocked to prevent shifting) to a government furnished 2½ ton, M-35 truck and transported over the specially prepared course described in 4.9.1 for 10 laps at a speed of 5 mph, 10 laps at a speed of 10 mph, 10 laps at a speed of 15 mph and 10 laps at a speed of 18-20 mph. A lap is defined as traversing the course in one direction. The facility shall meet the requirements of paragraph 3.5 after completion of the test.

NOTE: The facility shall be loaded to simulate the AN/TCC-73( )(V)1 using live and dummy loads. (See 6.2b for this test).

4.9.1 Vehicular transportation course.- The course for the military truck transportation shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1-inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be anchored securely by any convenient means. The ground on which the timbers are placed shall be hard and level.

4.10 Rail transport, impact test.- This test shall be performed on a level stretch of track in the manner described in this paragraph. The shelter facility shall be loaded as in 4.9. The test flatcar, timber blocking, and shelter facility tiedown strapping shall be as specified in Figure 3 and shall meet the provisions of the test as specified in 4.10.1; the test flatcar, less the shelter facility, and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in 4.10.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the government)

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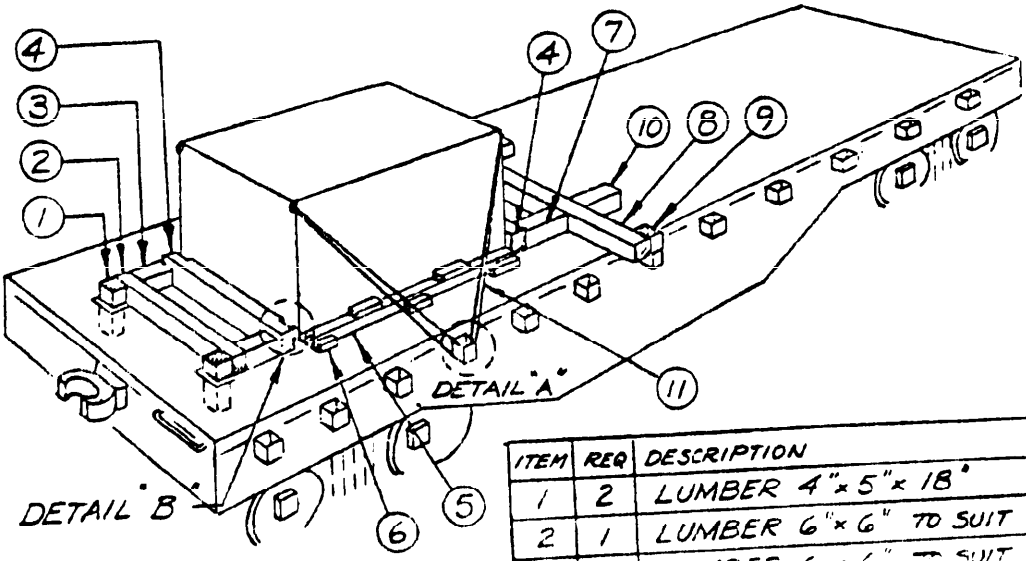
shall be impacted against the stationary loaded testcar coupled to the two empty flatcars, all with brakes off. This test shall be conducted four times; the first two runs shall be with the door-end of the shelter facility facing the loaded gondola car; the second two runs shall be with the door end of the shelter facility facing the opposite direction from the loaded gondola car. The lashing, blocking, and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during the impact, the run shall be repeated. After each impact, the shelter facility shall meet the requirements of 3.6.

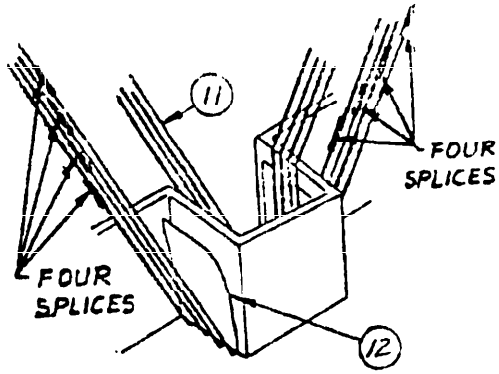
4.10.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch and 8 x 8 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The 8 x 8 inch blocking shall be notched to provide ¼ inch clearance around the towing eye castings of the shelter; an oak wedge shall be installed into this clearance as shown in the figure referred to above at each of four corners of the shelter. The 4 x 4 inch and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the figure referred to above. All blocking around the shelter facility shall be snugly against the shelter. (all other blocking shall be snugly abutting as shown in the figure.) The tiedown strapping shall be installed using a strapping machine U.S. Steel Corp., Model GT, or equal. If any tiedown strapping loosens or breaks during any run, it shall be replaced before the next run using new wire and a strapping machine as specified above.

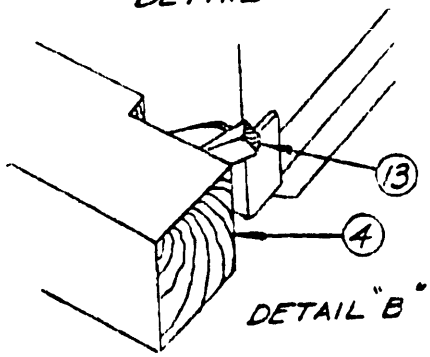
4.10.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, crushed concrete, metal scrap other than sheet metal or metal shavings or other springy forms of metal.

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ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x 5"x 18"
2	1	LUMBER 6"x 6" TO SUIT
3	2	LUMBER 6 x 6" TO SUIT
4	2	LUMBER 8 x 8" TO SUIT
5	2	LUMBER 4"x 4" x 147"
6	10	LUMBER 2"x 4" x 14"
7	2	LUMBER 6"x 6" TO SUIT
8	1	LUMBER 6"x 6" TO SUIT
9	2	LUMBER TO SUIT
10	2	LUMBER 6"x 6"x 24"
11	16 PER SHEET	ROUND STEEL STRAPPING #8 W&M GA.(.162" DIA.) U.S. STEEL CO. GR.C, OR EQ'L. 100,000 P.S.I. ULT. TENS. STR. 4 SEPARATE LOOPS EA. CORNER
12	4	FLEX. METAL SHEETING
13	4	OAK WEDGES, 1" WIDE

FIGURE 3 - SHELTER SECURED ON FLATCAR

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b. Chunks, slab, block, or plate form of the preceding acceptable materials, constrained to prevent shifting during impact.

4.11 Drop tests.-

4.11.1 Flat drop.- The shelter facility loaded as in 4.9 shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of 3.7.

4.11.2 Rotational drop.- The shelter facility loaded as in 4.9 shall have one skid resting on a nominal 4 inch high timber. The outer edge of the opposite skid shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four (4) times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of 3.7.

4.12 Watertightness test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc. shall be subjected to the following two part test: (Note. Perform tests with inside covers of entrance boxes removed).

a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Company, Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the

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roof nozzles which shall be at a distance of thirty-two inches. All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See Figure 4 for nozzle location). Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of 4.13, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door-end panel). Upon completion of the above test, the shelter facility shall meet the requirements of 3.8. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

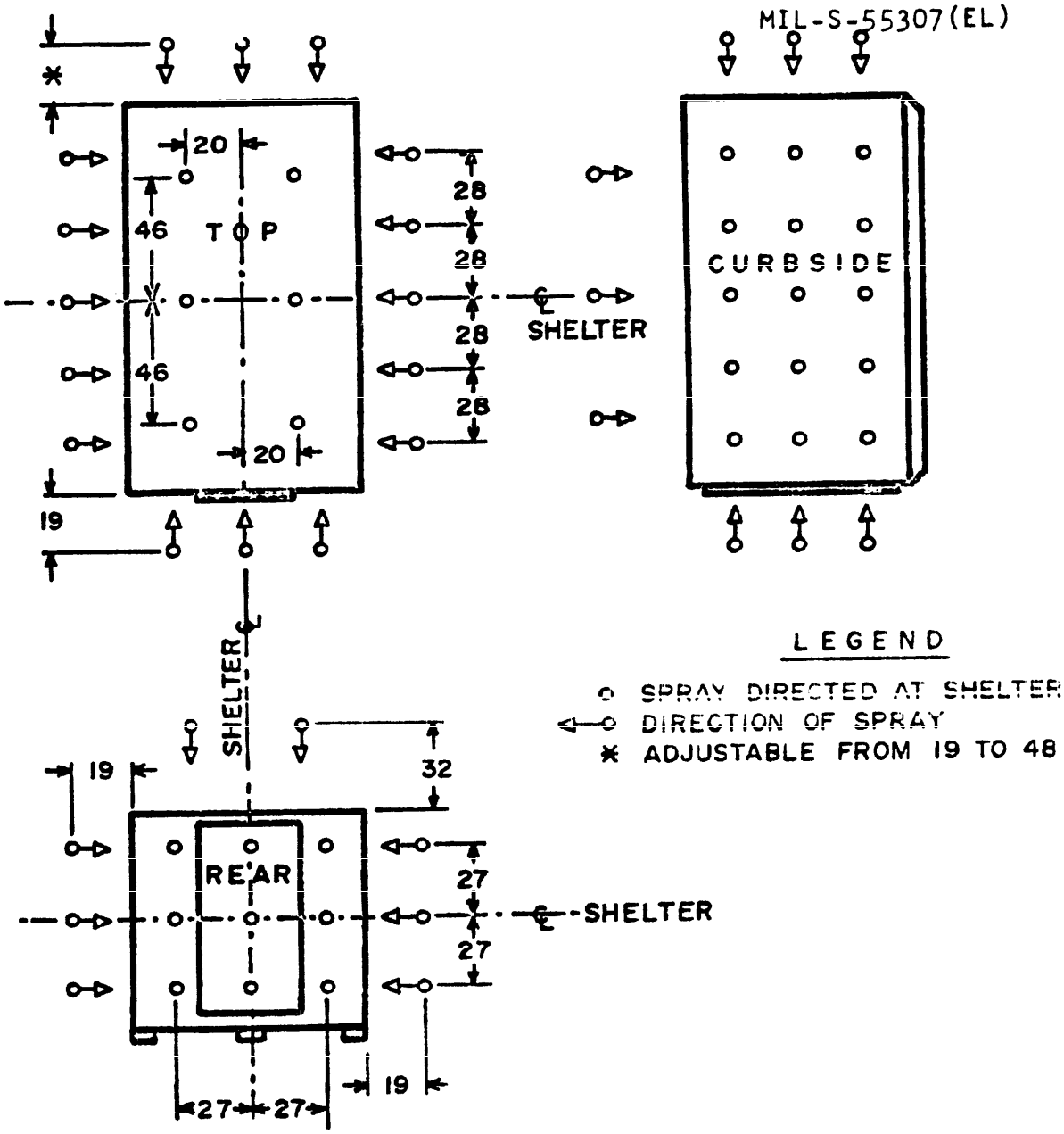
b. Open all covers to operating position (with exception of the door vent panel at the rear of the shelter and the air conditioner duct ports) and repeat a above.

4.13 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 21 inches (measured from the bottom of the skids) and meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tiedowns. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.14 Mounting insert tests.-

4.14.1 Preproduction and Group C insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

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NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 4
PAGE 55

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4.14.2 Production Group B insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

Table V - Mounting inserts loads

Insert	Group C and Preproduction test		Group B Production Test	
	Axial(lbs)	Torque(in-lbs)	(Axial(lbs)	Torque(in lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

4.15 Testing of solderless terminal-lug-connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.20.4.

4.16 Inspection for interchanbeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

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b. Size and mounting dimensions of racks, patch panels, and cabinets.

4.17 Visual and mechanical inspection.- The shelter facility and major subassemblies such as racks, entrance boxes, patch panels and blowers (prior to their installation into the shelter facility) shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table VI.

Table VI - Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated. 2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible. 3. Welds - cracked or porous. 4. Sealer improperly applied (holes, separations, or lack of adhesion). 5. Mounting inserts missing or mislocated. 6. Burrs or sharp edges in wire duct not removed. 7. Equipments or bracketry loose or missing. 8. Doors, etc. inoperative. 9. Electrical receptacles, switches, jacks, connections, wire ducts or other electrical components loose, improperly located or inoperative. 10. Locking or holding devices missing or inoperative. 11. Gaps between sections of wire duct exceed 1/16".

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Table VI - Classification of Visual and Mechanical Defects (cont'd)

Classification	Defects
Minor	1. Doors not easily operable. 2. Fastening devices difficult to operate. 3. Finish - abrasions or scratches

4.18 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.19 Rough handling test (preparation for delivery.- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.7 Electrical installation

Paragraph 4.17 Visual and mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the shelter facility shall be in accordance with the applicable requirements of Specification MIL-S-55505.

6. NOTES

6.1 Intended use.- The Shelter, Electrical Equipment S-419()/TCC-73(V) is the shelter facility portion of the AN/TCC-73()(V)$_1$ or AN/TCC-73()(V)$_2$ and it is used to house the operational equipments including:

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a. 8 ea CV-1548()/U Converter, Telephone Signal

b. 8 ea TD-660()/G Multiplexer

c. 4 ea TD-204()/G Multiplexer or TD-754()/G Multiplexer

d. 2 ea TD-976()/G Multiplexer

e. 2 ea H-91()/U Headset-Microphone

f. 2 ea H-156()/U Handset

g. 1 ea TA-312()/PT Telephone

h. 1 ea LS-147()/FI Intercom

i. 8 ea Security Equipment TSEC/KG-27() (Optional) forming the telephone terminal.

NOTE: The AN/TCC-73()$(V)_2$ contains the same equipment less the two TD-976()/G Multiplexers.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Preproduction and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads for the government furnished live loads (radio components) unless otherwise specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-419()/TCC-73(V).

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d. A list of equipments to be government loaned for testing purposes.

e. Submission of the statement referenced in 3.18.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally non-destructive determination of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determination of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.5).

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-419R/TCC-73(V). The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.6 Verification inspection.-Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army (EL)

Preparing Activity:
Army -EL

Project No. 5805-A162

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Circuit No. 7 - (Main circuit breaker CB-1)

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Shelter, Electrical Equipment S-419()/TCC-73(V)

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.
ESC-FM 1068-68

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Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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Commanding General
U.S.Army Electronics Command
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